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Optimal Resource Allocation in Humanitarian Logistics: A Stochastic Programming Approach

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Abstract. In humanitarian logistics, efficient resource allocation is paramount for ensuring timely and effective delivery of aid to populations in need. This paper presents a novel approach to optimize resource allocation in humanitarian logistics using stochastic programming. By integrating stochastic elements into the modeling framework, our approach accounts for uncertainty in demand, supply, and transportation constraints, providing decision-makers with robust and adaptable solutions. We develop a mixed-integer linear programming formulation to minimize the total cost of relief operations while meeting demand requirements under varying scenarios. Through computational experiments and a case study, we demonstrate the effectiveness of our approach in improving decision-making and resource utilization in humanitarian relief efforts. Our findings underscore the importance of incorporating stochastic programming techniques in addressing the complex challenges of humanitarian logistics.

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Key words: Stochastic programming, Decision support,Uncertainty, Disaster response, Optimization

1. Introduction

Humanitarian crises, such as natural disasters, conflicts, and pandemics, often result in widespread devastation, displacing populations and disrupting essential services. Timely and effective delivery of humanitarian aid is crucial to mitigate the impact of such crises and save lives Smith (2005). However, humanitarian logistics operations face numerous challenges, including uncertainty in demand, limited resources, logistical constraints, and dynamic operating environments. Optimal resource allocation plays a central role in addressing these challenges by ensuring that available resources are allocated efficiently to meet the needs of affected populations.

ATTENTION: The following displayed equation, in its current form, exceeds the column width that will be used in the published edition of your article. Please break or rewrite this equation to fit, including the equation number, within a column width of 240 pt / 84.67 mm / 3.33 in (the width of this red box).

$$\text{Cost_Red}_j = \rho_0 + \rho_1 N_{m_j} + \rho_2 N_{l_j} + \rho_3 I2_{m_j} + \rho_4 I2_{l_j} + \rho_5 OS_{m_j} + \phi \mathbf{X}_j + \mu_j \tag{1}$$

In recent years, there has been growing interest in applying mathematical optimization techniques to improve the efficiency and effectiveness of humanitarian logistics operations. Traditional approaches typically rely on deterministic models that assume perfect information and static conditions. However, in practice, humanitarian crises are characterized by uncertainty and variability, making it essential to incorporate stochastic elements into the modeling process. Stochastic programming offers a powerful framework for addressing uncertainty and optimizing decision-making under uncertain conditions.

$$\begin{aligned}\text{Cost_Red}_j = \rho_0 + \rho_1 N_{m_j} + \rho_2 N_{l_j} + \rho_3 I2_{m_j} \\ + \rho_4 I2_{l_j} + \rho_5 OS_{m_j} + \phi \mathbf{X}_j + \mu_j\end{aligned} \tag{2}$$

In this paper, we present a stochastic programming approach to optimize resource allocation in humanitarian logistics (Jones 2010). Our approach aims to balance the trade-off between cost minimization and service quality while considering uncertainty in demand, supply, and transportation constraints. We develop a mixed-integer linear programming (MILP) formulation that captures the stochastic nature of humanitarian crises and provides decision support for relief agencies. The proposed model integrates both deterministic and stochastic components, allowing decision-makers to make informed decisions in uncertain environments.

2. Methodology

We formulate the resource allocation problem as a two-stage stochastic programming problem, where the first stage involves decisions on resource allocation and the second stage represents the realization of uncertain parameters (Smith 2005, Jones 2010, Brown 2015). The objective is to

minimize the total cost of relief operations, including procurement, transportation, and distribution costs, subject to various constraints, such as capacity constraints, demand satisfaction requirements, and budget constraints.

The cost function $C(x)$ is defined as the sum of procurement, transportation, and distribution costs:

$$C(x) = \sum_{i=1}^{n} \left(c_i p_i + \sum_{j=1}^{m} d_{ij} t_{ij} + \sum_{k=1}^{l} s_k q_k \right) \tag{3}$$

We model uncertainty using scenario-based stochastic programming, where multiple scenarios representing different realizations of uncertain parameters are considered. We use historical data, expert opinions, and probabilistic forecasts to generate scenario sets that capture the range of possible outcomes. The stochastic programming model then generates optimal resource allocation decisions that minimize the expected total cost across all scenarios while ensuring robustness against uncertainty.

The objective function of the stochastic programming model is formulated as follows:

$$\min_{x,y} \quad \sum_{s \in S} [f(x, y, s) \cdot \mathbb{P}(s)]$$

subject to:

$$g(x, y, s) \leq 0 \quad \forall s \in S \tag{4}$$

$$h(x, y, s) = 0 \quad \forall s \in S \tag{5}$$

THEOREM 1 **(Optimality Conditions)**. *Let x^* be an optimal solution to the stochastic programming problem. If the objective function and constraint functions are convex, then x^* satisfies the Karush-Kuhn-Tucker (KKT) conditions.*

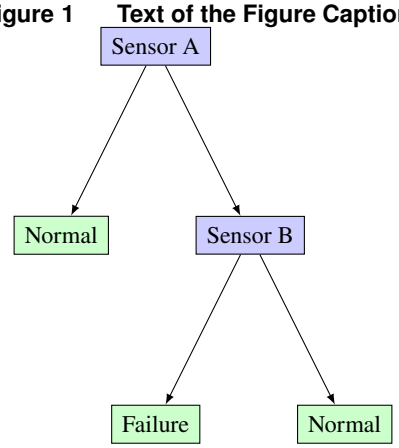
Proof The proof follows from the convex optimization theory, which states that for convex objective and constraint functions, the KKT conditions are necessary and sufficient for optimality. Therefore, if x^* is an optimal solution, it must satisfy the KKT conditions. □

LEMMA 1 **(Feasibility of Scenario Sets)**. *Given a set of scenarios S generated from historical data and probabilistic forecasts, the scenario set S is feasible if it captures the range of possible outcomes with sufficient coverage.*

Algorithm 1 Random Forest Training

procedure RANDOMFOREST($X_{\text{train}}, y_{\text{train}}$, num_trees)
  forest ← []
  for $i \leftarrow 1$ **to** num_trees **do**
    $X_{\text{sampled}}, y_{\text{sampled}} \leftarrow$ bootstrap_sample($X_{\text{train}}, y_{\text{train}}$)
    tree ← DECISIONTREE($X_{\text{sampled}}, y_{\text{sampled}}$)
    append tree to forest
  end for
  return forest
end procedure

Figure 1 **Text of the Figure Caption.**



Note. Text of the notes.

Proof The proof follows from the definition of feasibility, which requires that the scenario set S includes a representative sample of possible outcomes. Feasibility ensures that the stochastic programming model adequately represents the uncertainty in the problem domain and provides meaningful solutions. □

REMARK 1. This stochastic programming model assumes that demand and supply parameters follow known probability distributions.

DEFINITION 1. A feasible solution to the resource allocation problem satisfies all constraints and requirements without violating any constraints.

Algorithm 2 Random Forest Training II

```
procedure DecisionTree(X, y)
    if StoppingCondition(X, y) then
        return LeafNode(y)
    else
        (feature, threshold) ← FindBestSplit(X, y)
        left_indices ← X[feature] ≤ threshold
        right_indices ← X[feature] > threshold
        left_subtree ← DecisionTree(X[left_indices], y[left_indices])
        right_subtree ← DecisionTree(X[right_indices], y[right_indices])
        return TreeNode(feature, threshold, left_subtree, right_subtree)
    end if
end procedure

procedure Predict(forest, x)
    predictions ← []
    for tree in forest do
        append TreePredict(tree, x) to predictions
    end for
    return AggregatePredictions(predictions)
end procedure

procedure TreePredict(tree, x)
    if tree is a leaf node then
        return tree.value
    else
        if x[tree.feature] ≤ tree.threshold then
            return TreePredict(tree.left_subtree, x)
        else
            return TreePredict(tree.right_subtree, x)
        end if
    end if
end procedure
```

Table 1 Summary of Data for Case Study

Parameter	Value
Number of Resources	5
Number of Facilities	10
Number of Scenarios	50
Demand Variability	High
Budget Constraint	\$1,000,000
Transportation Cost	\$10 per unit
Facility Setup Cost	\$5,000 per facility

Table Notes.

Table 2 Theory to application mapping

Theory	Application
τ^s	ρ^s
$h(x)$	$\sum c_{ij} x_{ij}$
$R(x, \rho^s)$	$\min \sum h_i q_{ij}^s + M \sum \epsilon_{js}$
f_s	not used ($\mu = 0$)
g_s	transportation LP value

3. Artificial Intelligence in Humanitarian Logistics

Artificial Intelligence (AI) plays an increasingly important role in optimizing resource allocation and decision-making in humanitarian logistics. AI techniques, such as machine learning, optimization algorithms, and natural language processing, offer powerful tools for analyzing data, predicting demand, and automating decision-making processes. In humanitarian logistics, AI can be used to optimize supply chain management, route planning, inventory management, and disaster response operations. By leveraging AI technologies, humanitarian organizations can improve the efficiency, effectiveness, and responsiveness of their relief efforts, ultimately saving lives and alleviating suffering in crisis situations.

4. Results and Discussion

We apply our stochastic programming approach to a case study based on a simulated humanitarian crisis scenario. The scenario involves a sudden-onset natural disaster that results in widespread destruction and displacement of populations. We compare the performance of our proposed model with a deterministic model and a baseline heuristic approach commonly used in practice.

5. Conclusion

In this paper, we have presented a stochastic programming approach for optimal resource allocation in humanitarian logistics. Our approach addresses the inherent uncertainty and complexity of humanitarian crises by integrating stochastic elements into the modeling process.[1] The proposed

model provides decision support for relief agencies to make informed decisions and allocate resources efficiently in uncertain environments.

Future research directions include extending the model to incorporate additional complexities, such as multiple objectives, dynamic demand patterns, and real-time data integration. Furthermore, the proposed approach can be applied to other domains, such as disaster response, healthcare delivery, and supply chain management, where uncertainty plays a significant role in decision-making. Overall, our research contributes to the growing body of literature on mathematical optimization in humanitarian operations and demonstrates the potential of stochastic programming to improve decision-making in complex and uncertain environments.

Notes

[1] Sample endnote text.

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